

Foundations of Cyber Security

COMSM 0118 - Applied Cryptography

Cryptography

What is Cryptography?

*There is no such thing as **ART**, really.*

*There are only **ARTISTS!*** The story of Art. E. H. Gombrich

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There course is a series of short stories about what cryptographers do and how they see the world.

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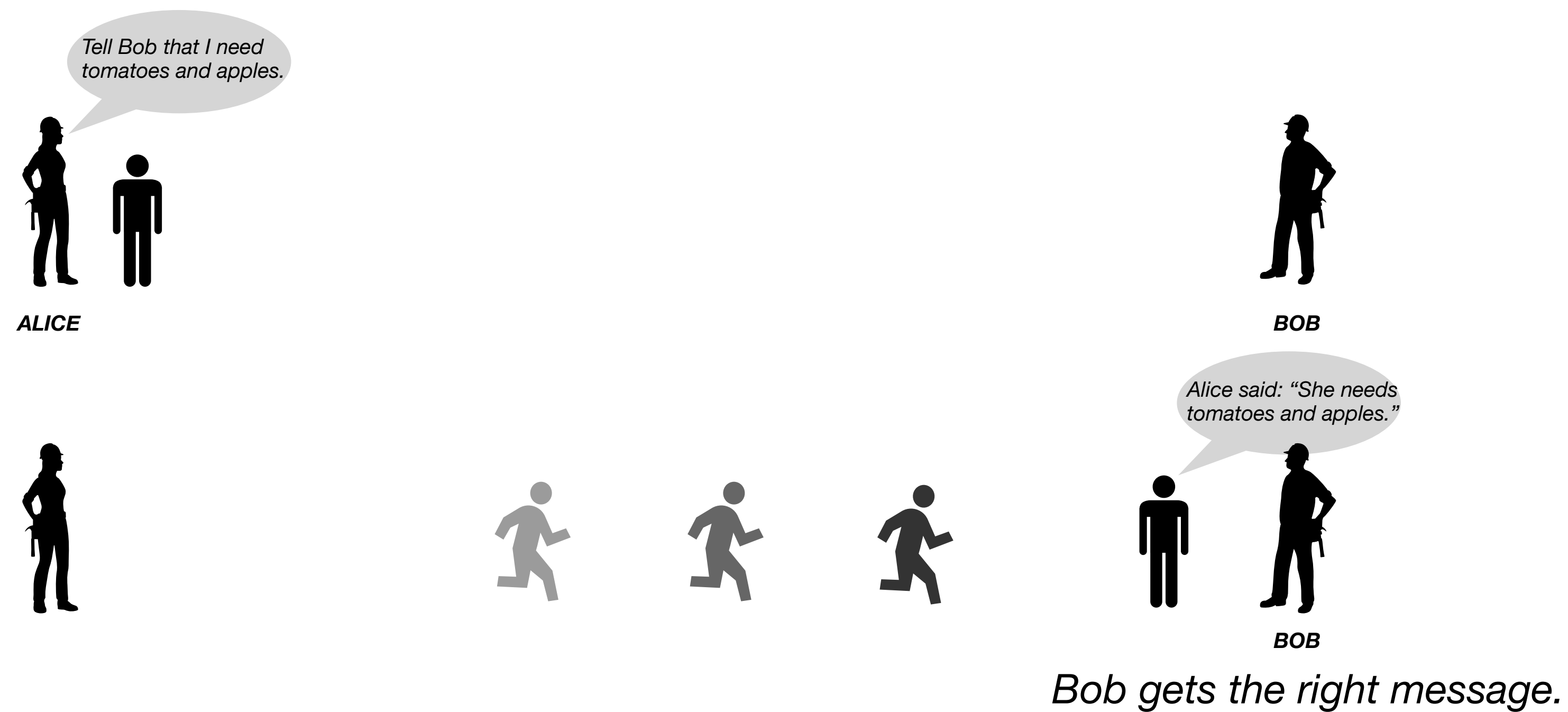
*What does it mean to say that **there is no such thing as art / cryptography, really; there are only artists / cryptographers?***

Cryptography

What is Cryptography?

To understand crypto, we first need to understand how two parties who want to share a secret message communicate.

In the early days, people used messengers (people, not applications) to relay their messages to people in far places like a war zone. Before writing was common, Person A would simply tell the messenger the information they wanted relayed to Person B. The messenger would find the person B and tell relay the message.



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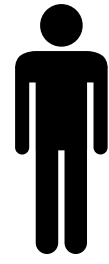


Bob gets the right message.

Cryptography

What is Cryptography?

Tell Bob that I need tomatoes,
ginger, cooking oil, and apples.
He should also bring in some
£10 cash with him.



ALICE



BOB

Alice said: "She needs
tomatoes, oil, and
apples. Bring cash also"

Security goal: *Authenticity*

*Bob does not get an accurate
message. Some bits of the
message have been omitted.*

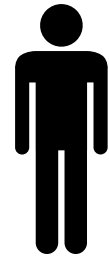
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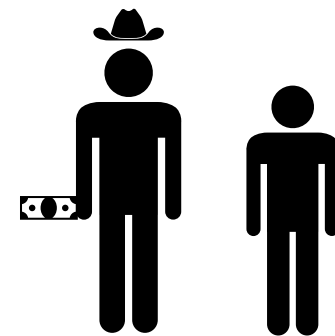
Security goal: Authenticity

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ALICE

Can you go to Bob and tell him that Alice says she needs £10 for fish and chips.



MALLORY



Alice said she needs £10 for fish and chips.



BOB

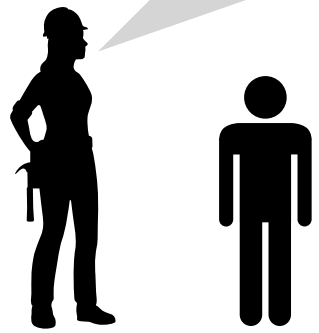
Security goal: Identity

Mallory pretends to be Alice to get money from Bob.

Cryptography

What is Cryptography?

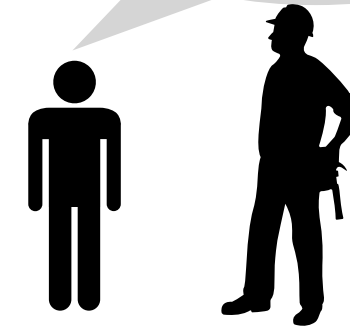
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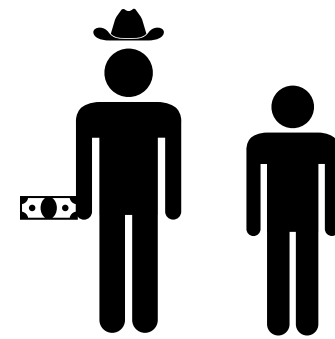
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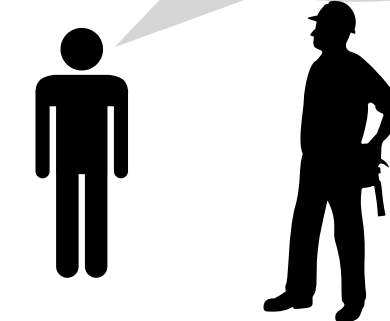
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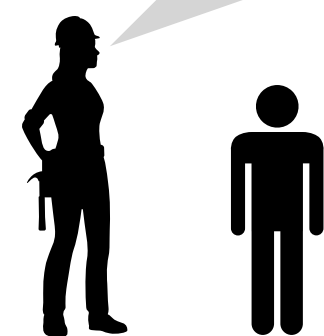


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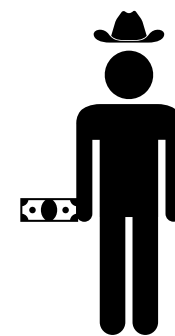
Mallory pretends to be Alice to get money from Bob.

Tell Bob that he should not come home, the woman we owe some money is here.



ALICE

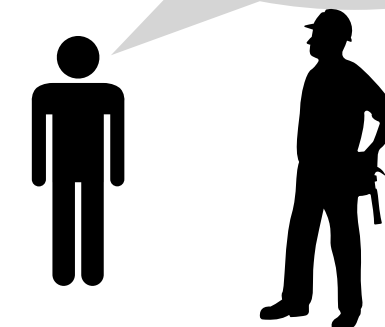
Change the message and tell Bob to come home soon with some money.



MALLORY



Alice said: "Come home soon and bring some money with you"



BOB

Security goal: Authenticity

Bob gets the wrong message. Mallory asked the messenger to change the message.

Cryptography

What is Cryptography?



ALICE



BOB

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F

A → G

D → J

L → R

Cryptography

What is Cryptography?



A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
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A → G
D → J
L → R

LUNCH IS READY.



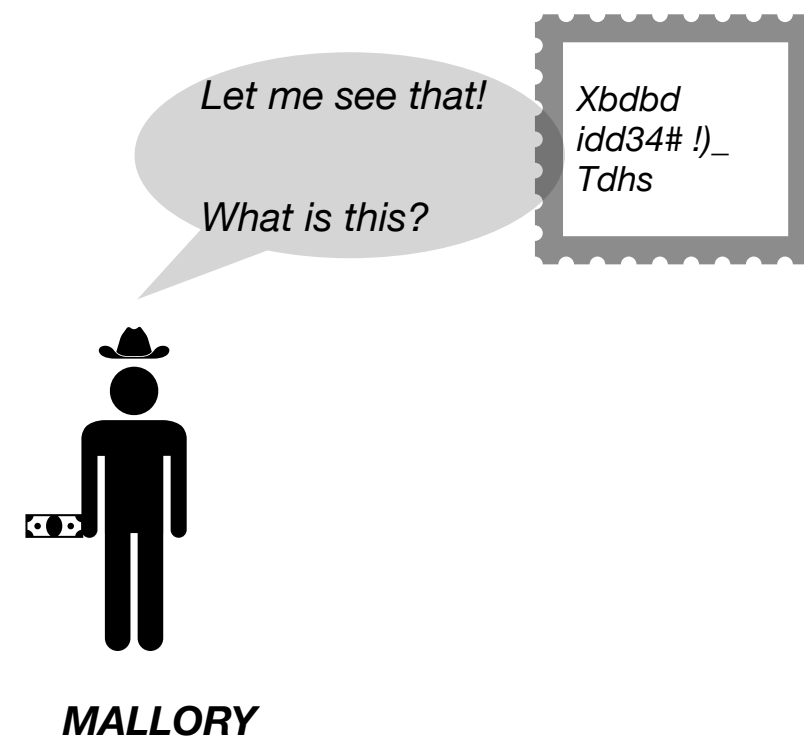
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Cryptography

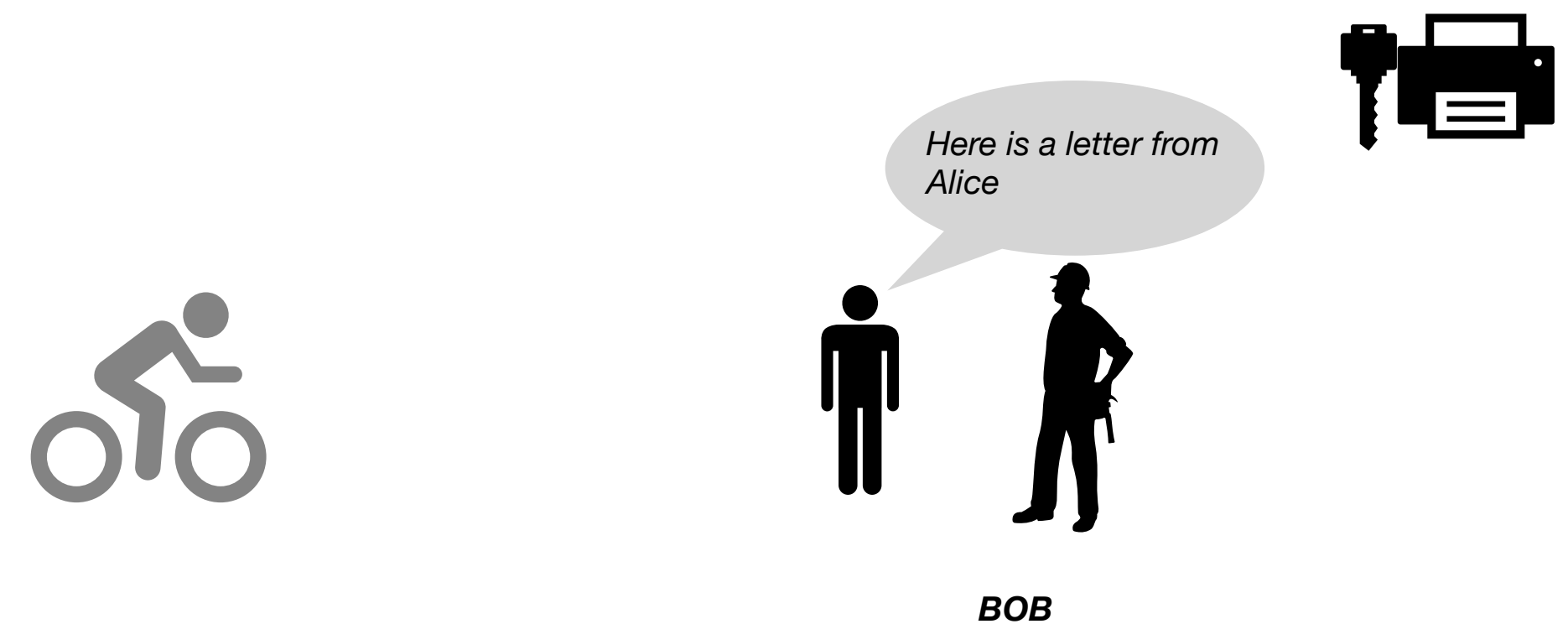
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Alice writes a message but in a form that can only be understood by Bob.



As usual Mallory bribes the messenger to see the message. Though he can see the message, Mallory cannot understand it.



Bob gets the message, interpret it and he can confirm it is from Alice.

Cryptography is the practice of transforming information so that it cannot be understood by unauthorised parties.

Cryptography

What is Cryptography?



An adversary as a well-funded, technically capable and highly motivated entity with lots of computing power.

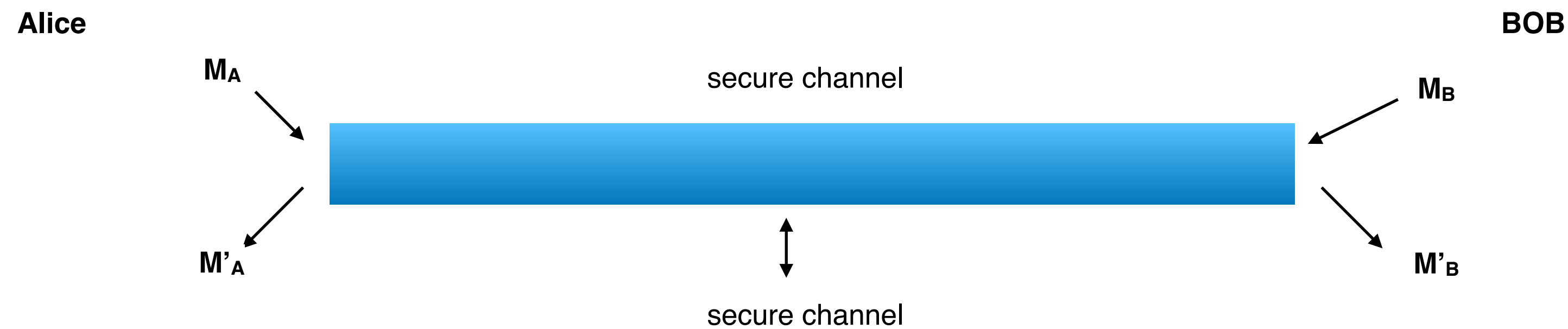
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The basic tasks of cryptography is establishing a secure channel. We want to do it and do it right.



Privacy: Adversary does not learn anything about M_A and M_B

Authenticity: $M'_A = M_A$ and $M'_B = M_B$

Cryptography

What is Cryptography?

“the practice of transforming information so that it cannot be understood by unauthorised parties.”

Cryptographic mechanisms to render messages meaningless to an adversary are broadly known as **Encryption**.

“I am fairly familiar with all forms of secret writings, and am myself the author of a trifling monograph upon the subject, in which I analyse one hundred and sixty separate ciphers, but I confess that this is entirely new to me.” *The Adventure of the Dancing Men: Sherlock Holmes* by Sir Arthur Conan Doyle

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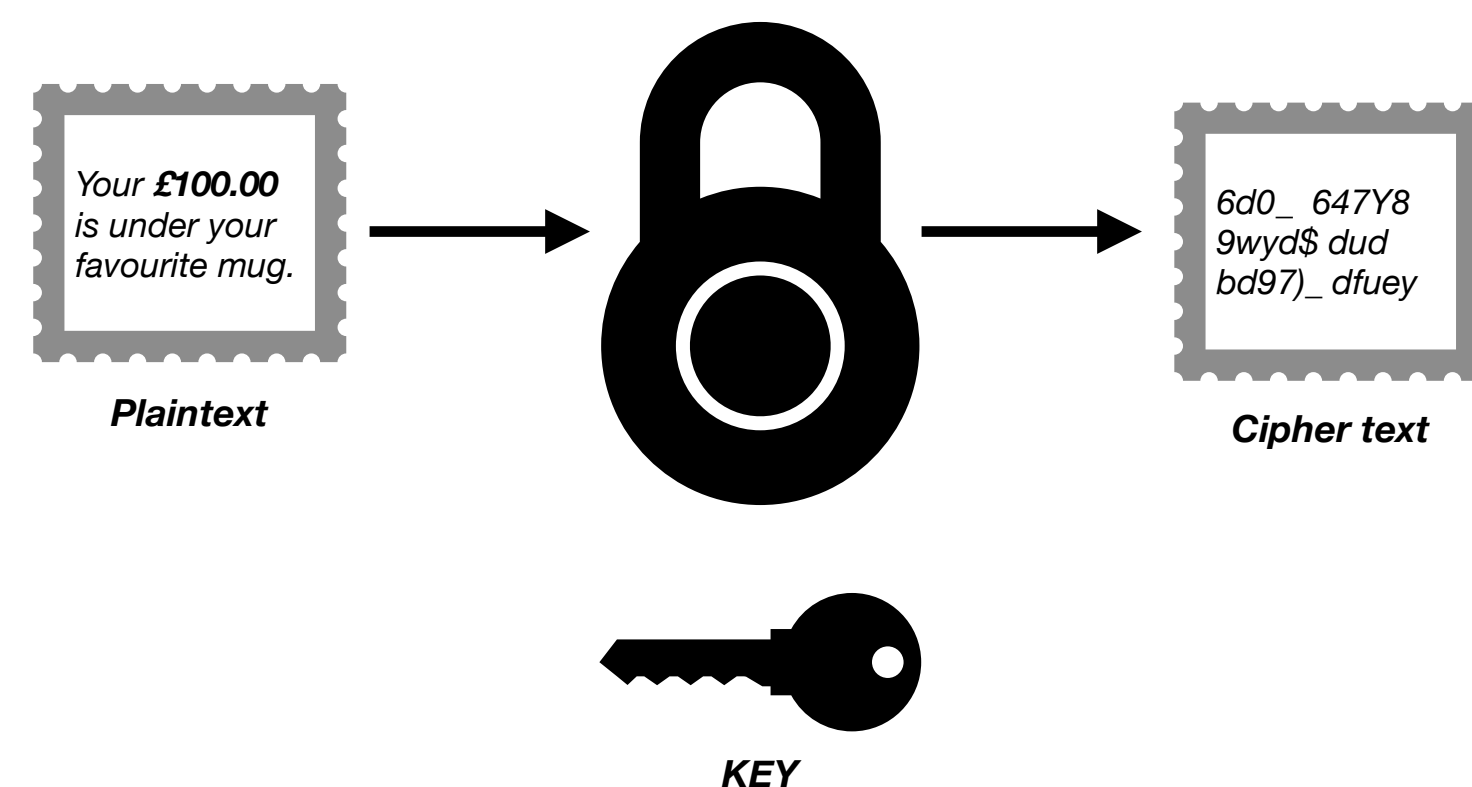
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Encryption

... the process of changing the original text into a scrambled or recognisable message.



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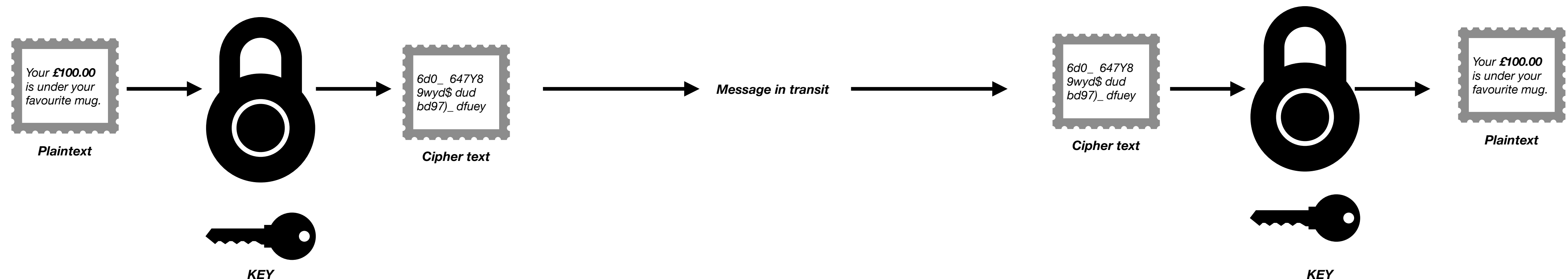
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Decryption

... the process of changing the scrambled or recognisable message back to its original form.



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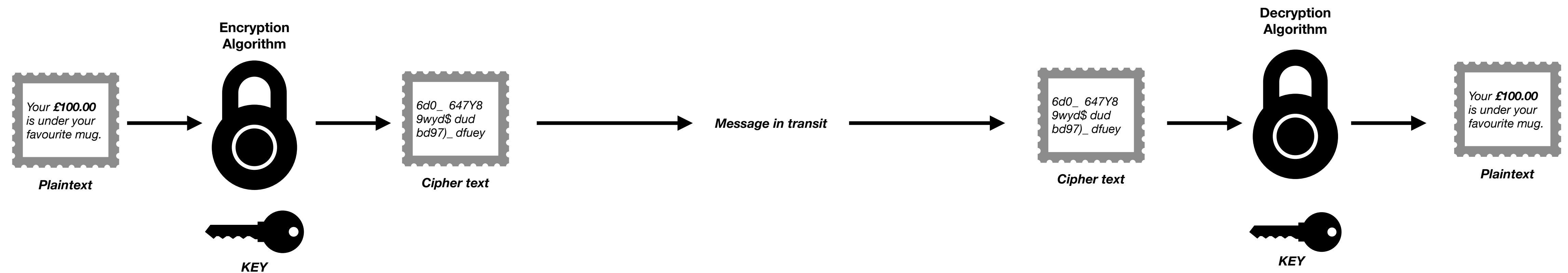
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Cryptographic Algorithm (also known as Cipher)

... mathematical procedures used to encrypt or decrypt data.

Key

... mathematical value or string of characters entered/input into a cryptographic algorithm to produce a ciphertext or plaintext.

Plaintext

... unencrypted data that is input for encryption or the out of the decrypted text.

Ciphertext

... is the scrambled or unreadable output (data) of the encryption.

Cleartext

... unencrypted data that is intended not to be encrypted.

Cryptography

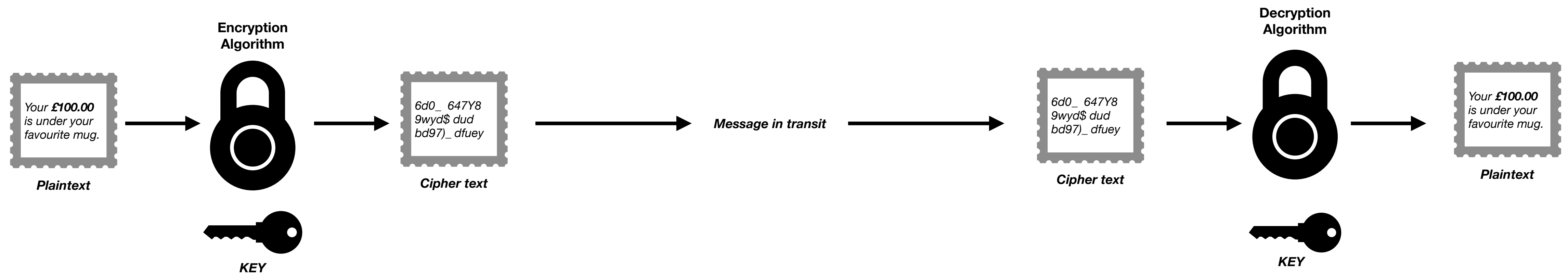
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Can be made
public!!!

Cryptographic Algorithm (also known as Cipher)

... mathematical procedures used to encrypt or decrypt data.

Should be
kept secret!!!

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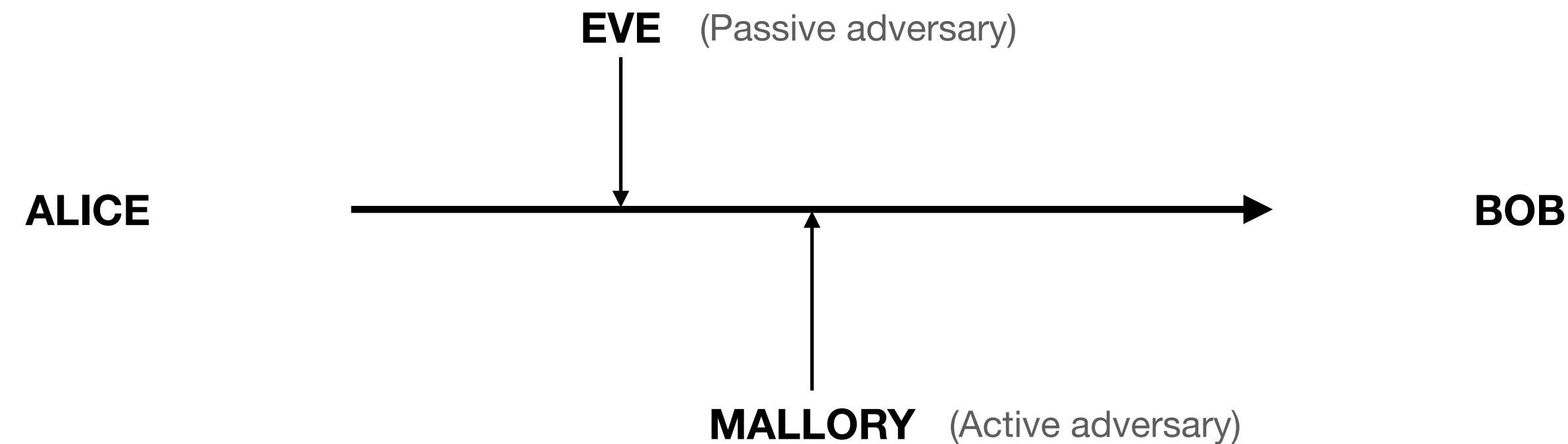
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Cryptography

Information Protection: Why use cryptography?

“Physics was good, more physics was even better. Cryptography is good, more crypto is even better...” Partha Das Chowdhury



Passive Adversary

An threat that can listen and understand all the communications sent between two parties.

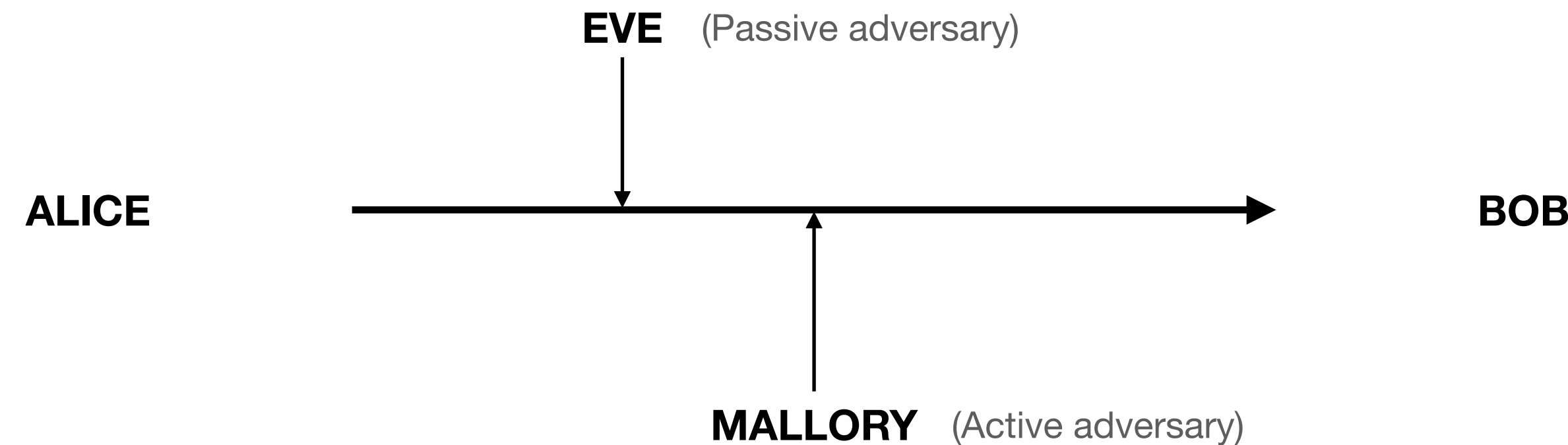
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An threat that can alter the communication sent between two parties or initiate communication pretending to be someone else.

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Confidentiality

It can help protect the confidentiality of the message from Alice to Bob, ensuring that neither Eve or Mallory can know the content of the message.

Integrity

It can help protect the integrity of the message from Alice to Bob. It can ensure that Mallory or any other unauthorised person (or software) can temper or alter the contents of the message during transmission. Without a cryptography key, Mallory cannot decrypt the message and change the content of the message.

Availability

It can protect the line of communication and ensure that communication and storage is always available.

Nonrepudiation

It can help to enforce nonrepudiation. Alice cannot deny that they sent the message. Repudiation is the act of denying. Nonrepudiation is the process of proving that someone (a user) has performed an action.

Masquerading

It can help to prevent users from pretending to be an authorised user or sender of the message. Mallory cannot pretend to be Alice.

Authentication

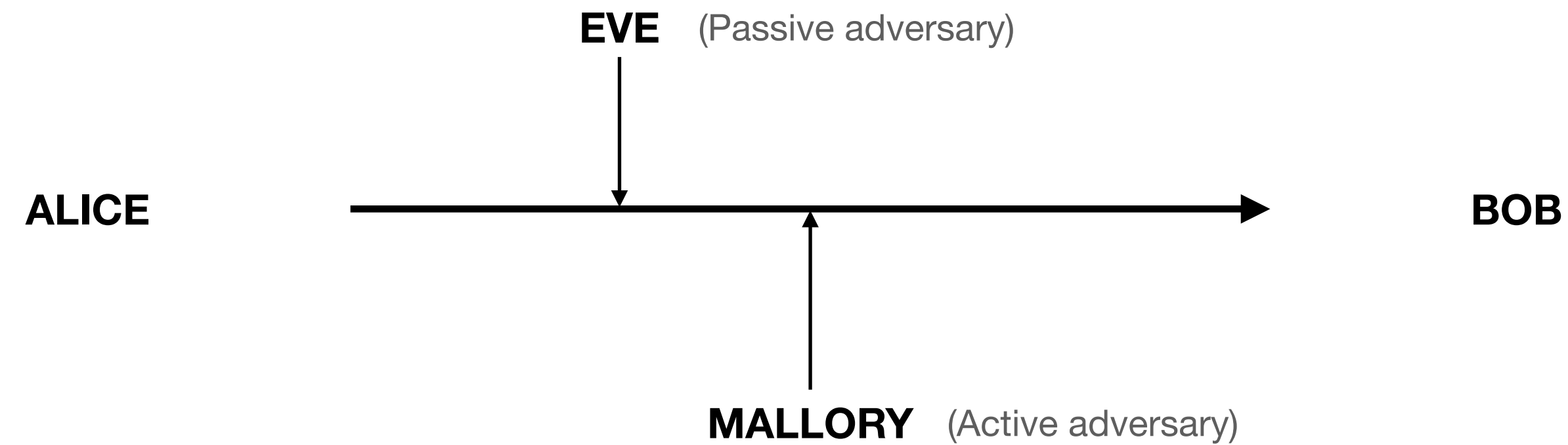
It can help us to verify the sender of the message. Bob can verify that the message was from Alice not Mallory.

Obfuscation

It can help us make the message appear to be meaningless. Crypto can help provide some level of obfuscation. Obfuscation is the action of making something obscure, unclear, or unintelligible.

Cryptography

Information Protection: Where?



Data in Processing

It can help protect data while being processed.

Data in transit

Crypto can help protect data between communicating parties.

Data at rest

It can help us to protect data stored on electronic devices/media.

Cryptography

History: Encryption Ciphers

Ciphers (algorithms) can be separated into several categories, with specific types in each category. One category is a **substitution cipher**, which exchanges one character for another. One old example is the *Caesar cipher*.

The Caesar cipher is a classic example of ancient cryptography and is said to have been used by Julius Caesar.

The Caesar cipher is based on transposition and involves *shifting* each letter of the **plaintext** message by a certain number of letters, historically three.

The **ciphertext** can be decrypted by applying the same number of shifts in the opposite direction.

Example

Shift by 6

A → G
D → J
L → R

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
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In this example, it is safe to say **6** is our **cryptographic key** as long as it gives us a letter that is part of the alphabets.

Cryptography

History: Encryption Ciphers

If we want to encrypt and decrypt message, we can mathematical show this by:

Basics

Let ***M*** denote the *Message* we want to encrypt.

Let ***K*** denote the *Cryptographic Key* we want to use.

Function to Encrypt the message can be presented by:

$$\mathbf{f(M,K) = M'}$$
 where M' is the ciphertext.

Function to Decrypt the message can be presented by:

$$\mathbf{f^{-1}(M',K) = M}$$

Caesar Shift Cipher

- Think of alphabets A to Z as numbers from 0 to 25.
- Think of another number as your secret, this is your key. $K \in \mathbb{Z}_{26}$
- We will encrypt message **M** with key **K** and produce a ciphertext **C**.
- Encrypt each plaintext character (M) as:

$$\mathbf{(M + K) \bmod 26}$$

to produce ciphertext C

- Decrypt each ciphertext (C) as:

$$\mathbf{(C - K) \bmod 26}$$

to produce plaintext M

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EXERCISE

Using the encryption function above, can you encrypt *plaintext* TEA where $K = 11$

If SPWWZ is a *Ciphertext* of HELLO, what is the value of K ?

Cryptography

History: Encryption Machines

A more intricate example of a cryptographic machine can be found in the German-made Enigma machine.

The Enigma was created by Arthur Scherbius in 1923 and was used to secure German communications during World War II.

In fact, there were several models of Enigma machine, and a variety of accessories and add-ons that could be attached to them.

The Enigma was based on a series of wheels, referred to as rotors, each with 26 letters and 26 electrical contacts on them, similar in general concept to the Jefferson Disk.

The device also had a keyboard, on which the plaintext message was entered, and a set of 26 characters above the keyboard, each of which could be lit.

To add further possible variations, some models also had a patch panel, allowing some or all the letters to be swapped by plugging cables into different positions.

On each rotor, the ring containing the letters of the alphabet could also be rotated independently of the electrical contacts, in order to change the relationship between the character selected and the character output.



EVOLUTION OF ENCRYPTION

Cryptography

History: Evolution of Encryption

The problem with the ciphers was that it was easy to break. When you understood the algorithm then it was easy to guess the key.



The secret behind the ciphers was the encryption algorithm itself. If the algorithm was compromised, then a new algorithm was to be developed.

Drawback: Compromised algorithm forces a definition of a new one (at high cost)

However, since the 70ies, the secret was no longer in the algorithm but the key, something external to the algorithm (secret = encryption key). This meant the algorithms could be made public.

Advantage: a compromised key forces changing the key, not the algorithm

Cryptography

History: Evolution of Encryption

In 1883, Dutch linguist and cryptographer Auguste Kerckhoffs published an article titled “La Cryptographic Militaire” (Military Cryptography) in the “Le Journal des Sciences Militaires” (Journal of Military Sciences). Principles that should be applied to a crypto system.

- *The system must be substantially, if not mathematically, undecipherable;*
- ***The system must not require secrecy and can be stolen by the enemy without causing trouble;***
- *It must be easy to communicate and retain the key without the aid of written notes, it must also be easy to change or modify the key at the discretion of the correspondents;*
- *The system ought to be compatible with telegraph communication;*
- *The system must be portable, and its use must not require more than one person;*
- *Finally, given the circumstances in which such system is applied, it must be easy to use and must neither stress the mind or require the knowledge of a long series of rules.*

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This principle states that ***the security of a cryptographic system must depend on the secrecy of its keys only and everything else, including the algorithm itself, should be considered public knowledge.***

Shannon's Maxim and the Open Design Principle

“One ought to design systems under the assumption that the enemy will immediately gain full familiarity with them.”

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*These two principles challenges the idea of **Security by Obscurity**. Security by Obscurity is the practice of keeping the system (or algorithm secret with the belief that it will never be public. Security by Obscurity is flawed since what makes it secure is the fact that it is unknown. The problem is you can't keep secret from everyone, eventually someone will know. This is why we obfuscation alone cannot be used to protect data. A lot of proprietary algorithms are weak because of this, they are not public therefore, not vetted thus contain flaws.*

Secure system in the light of Kerchoff's principle

An adversary knows the system or algorithm, that is, Function f and f^{-1} are public.

The security of the system should be dependent entirely on the key, that is, the key should be kept secret.